

TryHackMe: Couch Write-Up

Target Overview

- Target IP: 10.114.148.3
- Difficulty: Easy/Medium
- Objective: Obtain the `user.txt` and `root.txt` flags.

Phase 1: Enumeration & Reconnaissance

Network Scanning (Nmap)

We start by scanning the target machine to identify open ports and running services.

Bash

```
nmap -sC -sV -O 10.114.148.3
```

Scan Results:

Plaintext

PORT STATE SERVICE VERSION

22/tcp open ssh OpenSSH 7.2p2 Ubuntu 4ubuntu2.10 (Ubuntu Linux; protocol 2.0)

| ssh-hostkey:

| 2048 34:9d:39:09:34:30:4b:3d:a7:1e:df:eb:a3:b0:e5:aa (RSA)

| 256 a4:2e:ef:3a:84:5d:21:1b:b9:d4:26:13:a5:2d:df:19 (ECDSA)

|_ 256 e1:6d:4d:fd:c8:00:8e:86:c2:13:2d:c7:ad:85:13:9c (ED25519)

Aggressive OS guesses: Linux 3.8 - 3.16 (96%), Linux 4.4 (96%)

Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Web & Database Enumeration

By enumerating web services (specifically looking at CouchDB defaults, typically on port 5984 if exposed, or via a web proxy), we navigate to the CouchDB web administration interface:

- Path: `http://10.114.148.3/_utils/`

Inside the CouchDB dashboard, we explore the existing databases:

1. Located a database/directory named `/secret`.
2. Inspected the documents within `/secret` and discovered credentials for a user named `atena`, along with their password.

Phase 2: Exploitation & Initial Access

SSH Login

Using the discovered credentials, we connect to the target machine via SSH.

Bash

```
ssh atena@10.114.148.3
```

- **Username:** atena
- **Password:** *(Input the password found in the CouchDB database)*

Capturing the User Flag

Once logged in as atena, we can locate and read the user flag.

Bash

```
cat /home/atenaluser.txt
```

Phase 3: Privilege Escalation

Post-Exploitation Enumeration

To find a path to root, we check the user's command history for potential misconfigurations or clues left behind.

Bash

```
cat /.bash_history
```

Inside the history file, we uncover a specific docker command that the user has privileges to run.

Exploiting Docker for Root Access

By executing the custom Docker command found in the history, we are able to spawn a container that breaks out of the host restrictions or grants elevated privileges, effectively giving us a root shell.

Bash

```
# Run the specific docker command discovered in .bash_history
docker run [...]
```

Capturing the Root Flag

Now that we have root access, we search the file system for the final flag.

Bash

```
find / -type f -name "root.txt" 2>/dev/null
```

The search reveals the location of root.txt. We navigate to the directory and read the flag:

Bash

```
cat /mnt/root/root.txt
```

